Object Detection Extension

https://github.com/aiyshwary/Object-Detection-Extension-in-Rapid-Miner

Python PyTorch ONNX Runtime RapidMiner Object Detection

OVERVIEW

A RapidMiner Studio extension that brings transfer learning-based object detection into a no-code environment, supporting fine-tuning and inference with Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet models.

IMPACT

Developed two RapidMiner operators for object detection — FineTuneObjectDetectionModel and ObjectDetectionModelInference — enabling users to train and deploy detection models within a visual workflow without writing code.

TECHNICAL WALKTHROUGH

2. Object Detection Extension
Problem Statement
Different object detection use cases require different trade-offs between accuracy, speed, and computational cost. A single model cannot satisfy all scenarios.
Approach
Developed a unified object detection extension using PyTorch, supporting multiple architectures:
Faster R-CNN
FCOS
SSD
SSD Lite
RetinaNet
YOLO (extended work)
Training
Dataset annotated using XML bounding boxes
Transfer learning using COCO pre-trained weights
Fully configurable training pipeline:
Model selection
Batch size
Epochs
Learning rate
Momentum
Weight decay
Evaluation metrics:
IoU (Intersection over Union)
Precision
Recall (per class)
Inference
Users can choose any trained or pre-trained model
Output includes:
Image with bounding boxes, labels, and confidence scores

Tabular output with detected objects
Results also generated in base64 format for easy deployment and integration
Key Observations
Faster R-CNN performed best for accuracy
SSD and SSD Lite were faster but had lower recall
RetinaNet handled class imbalance well
Model choice depends on application requirements

KEY DETAILS

- FineTuneObjectDetectionModel accepts image and annotation directories (YOLO/Pascal VOC) and outputs model weights and class list CSV.
- ObjectDetectionModelInference runs predictions on new images and returns bounding boxes and class labels as a DataFrame.
- Supports five architectures: Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet for varying speed-accuracy trade-offs.
- Configurable for CPU-only and GPU (CUDA cu118) environments; installed as a JAR extension in RapidMiner Studio.
- Enables domain-specific object detection with minimal ML expertise using a drag-and-drop process designer.